Kores Date : Page : Sub :

Assignment 1

Q1. Complete the table with equivalents in Binary, octal, and Hexa

Decimal	Binary	Octal	Hexa
143.75	10001111.11	217.6	8F.C
27.625	11011.101	33.5	1B.A
299	100101011	453	12B
31.75	11111.11	37.6	1F.C

Q2 Find equivalents :

a) $324_5 = ?_{10}$

$\rightarrow 3\times5^2 + 2\times5^1 + 4\times5^0 = 75+10+4 = \underline{89_{10}}$

b) $32.23_4 = ?_{10}$

$\rightarrow 3\times4^1 + 2\times4^0 + 2\times4^{-1} + 3\times4^{-2} = 12+2+0.5+0.1875$

$\underline{14.6875_{10}}$

c) $231_{10} = ?_7$

$$\begin{array}{r|l} 7 & 231 \\ 7 & 33 - 0 \\ & 4 - 5 \end{array}$$

$\Rightarrow \underline{450_7}$

d) $189_{10} = ?_6$

$$\begin{array}{r|l} 6 & 189 \\ 6 & 31 - 3 \\ & 5 - 1 \end{array}$$

$\Rightarrow \underline{513_6}$

Q3. Find the radix r:

a) $(312/20)_r = (13.1)_r$

$3r^2+r+2 / 2r = r+3+1/r$

(x by $2r$)

$3r^2+r+2 = 2r^2+6r+2$

$r^2-5r=0$

$r(r-5)=0$

$r=5$

b) $264_r + 136_r = 433_r$

$(2r^2+6r+4)+(r^2+3r+6) = 4r^2+3r+3$

$3r^2+9r+10 = 4r^2+3r+3$

$r^2-6r-7=0$

$(r-7)(r+1)=0$

$r=7$

c) $221505_8 = 12345_r$

$2(8^5)+2(8^4)+1(8^3)+5(8^2)+5(8^0) = r^4+2r^3+3r^2+4r^1+5$

$74565 = 16^4+2(16^3)+3(16^2)+4(16^1)+5$

$74565 = 74565$

$r=16$

Q4. Addition

a)
$$\begin{array}{r} 00001101_2 \\ 1101111_2 \\ \hline 1111100 \end{array}$$

b)
$$\begin{array}{r} 7\;7\;6\;5\;2 \\ 7\;6\;5\;4\;3 \\ \hline 1\;7\;6\;4\;1\;5 \end{array}$$

$8|\underline{9}$ $1-1$ $\quad 8|\underline{12}$ $1-4$ $\quad 8|\underline{14}$ $1-6$ $\quad 8|\underline{15}$ $1-7$

c)
$$\begin{array}{r} 5\;7\;8\;A \\ A\;B\;D\;E \\ \hline 1\;0\;3\;6\;8 \end{array}$$

$16|\underline{24}$ $1-8$ $\quad 16|\underline{22}$ $1-6$ $\quad 16|\underline{19}$ $1-3$ $\quad 16|\underline{16}$ $1-0$

d)
$$\begin{array}{r} 2\;3\;4 \\ 3\;1\;5 \\ \hline 5\;5\;3 \end{array}$$

$6|\underline{9}$ $1-3$

Q5. Binary - Gray Code

a) 10101_2

$1 \oplus 0 \oplus 1 \oplus 0 \oplus 1$

$\rightarrow 1\;1\;1\;1\;1$

b) 01110_2

$0 \oplus 1 \oplus 1 \oplus 1 \oplus 0$

$\rightarrow 0\;1\;0\;0\;1$

Q6. Gray Code → Binary

a) 0 1 1 0 1 (gray)

0 1 1 0 1

0 1 0 0 1

b) 1 0 1 0 1

1 1 0 0 1